

# A Comparative Analysis of Models for Real-Time Personal Protective Equipment Detection on Edge Devices

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# Roadmap

1 of 1



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|                     |                                           |                                 |                    |                                |                               |                   |                    |
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| <b>Introduction</b> | <b>Indoor local<br/>PPE<br/>detection</b> | <b>Adopted<br/>architecture</b> | <b>Experiments</b> | <b>Dataset<br/>description</b> | <b>Model<br/>optimization</b> | <b>Conclusion</b> | <b>Future Work</b> |
|---------------------|-------------------------------------------|---------------------------------|--------------------|--------------------------------|-------------------------------|-------------------|--------------------|

# Introduction

## 1 of 3 – Problem statement



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47% of workplace accidents are related to incorrect use of personal protective equipment

| Type of error | % of incidents |
|---------------|----------------|
| Lack of PPE   | 34%            |
| Misuse        | 13%            |

URL: <https://ilostat.ilo.org/topics/safety-and-health-at-work/#>

| Areas to protect | Required PPE |
|------------------|--------------|
| Head             | Helmet       |
| Chest            | Vest         |
| Hand             | Glove        |

# Introduction

## 2 of 3 – PPE detection



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- **Human supervision** is prone to errors, costly, and cannot cover large areas;
- **Computer-based** supervision automatically detect dangerous situations:



**Sensor-based**



**Vision-based**

# Introduction

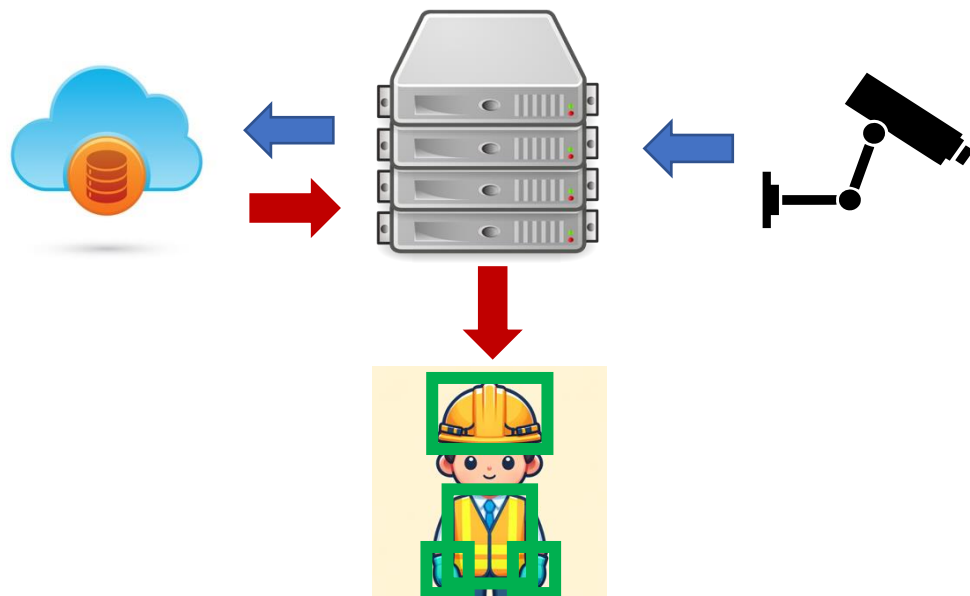
## 3 of 3 – Cloud vs Local Detection



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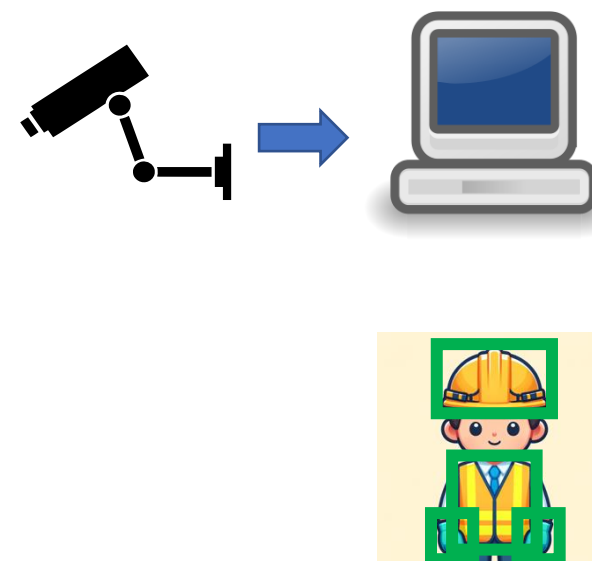
### Cloud-based system:

sends images through the network and waits for response



### Local-based system:

acquires images and detect objects on the same machine

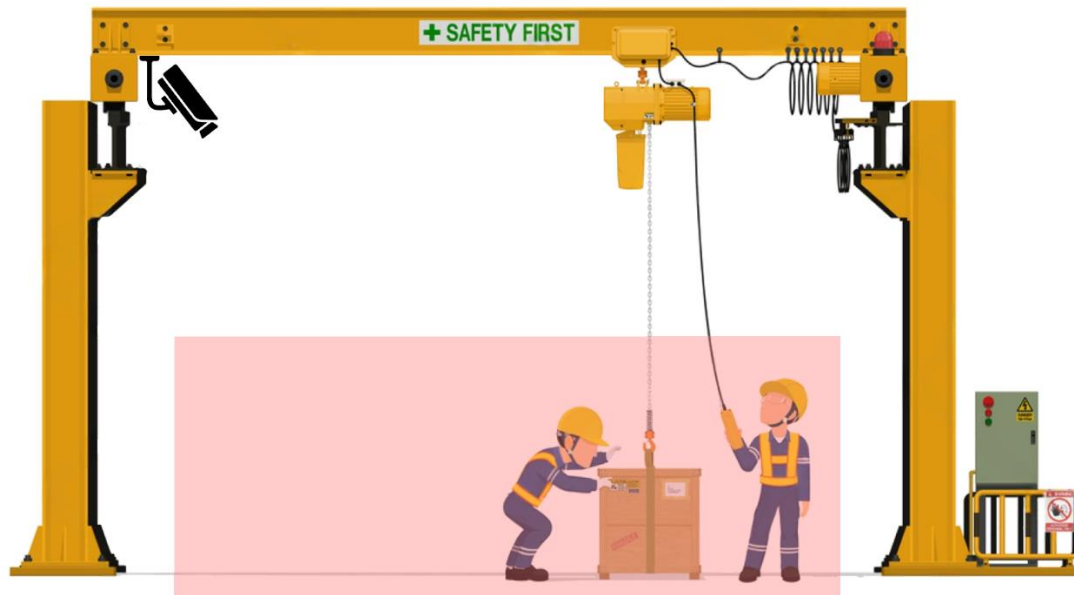


# Scenario: Indoor local PPE detection

## 1 of 1 – Crane-Zone PPE Detection



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- Overhead crane setup in indoor industrial facilities;
- Camera mounted on the crane enables wide-area monitoring from above;
- Detection focused on high-risk zones directly beneath the crane;
- Immediate alarm if PPE violation is detected (e.g., missing helmet or vest);

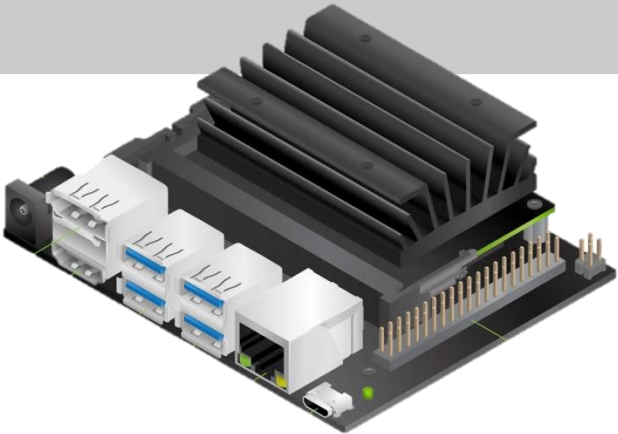
# Adopted Architecture

1 of 1



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| Camera                             | Edge device                              |
|------------------------------------|------------------------------------------|
| Intel RealSense D435 (RGB + depth) | NVIDIA Jetson Nano (4GB RAM, <10W power) |



| Input stream                   | AI inference                                    | Output                                                              |
|--------------------------------|-------------------------------------------------|---------------------------------------------------------------------|
| Real-time RGB and depth frames | On-device PPE detection using YOLOv11 / RT-DETR | Alerts triggered for PPE violations or proximity to hazardous zones |

### Objective

Evaluate real-time PPE detection models on NVIDIA Jetson Nano.

### Models Tested

- **YOLOv11** (Nano, Small, Medium, Large, XLarge)
- **RT-DETR** (L, X)

### Metrics

- **Accuracy:** AP@50, mAP
- **Real-Time Performance:** FPS
- **Hardware Metrics:** Disk Occupancy, Board Temperature, Power Consumption



## 2 of 2 – Workflow

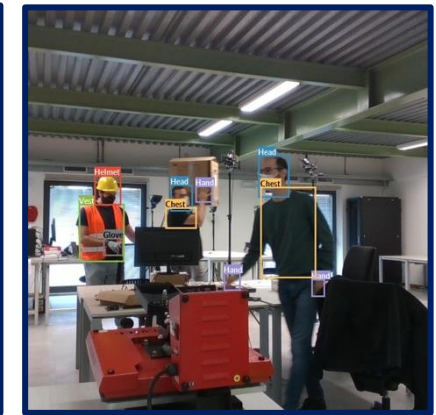
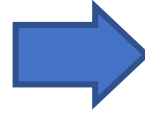
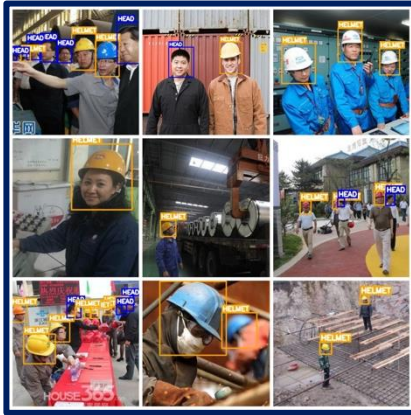
[illegible]

# Dataset description

1 of 1 – General Overview



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Public dataset

Public dataset  
cleaned and  
labeled

Bright-night  
dataset  
green screen

Laboratory  
dataset  
real scenario

7035 images

6832 images

+

215 images

+

236 images

=

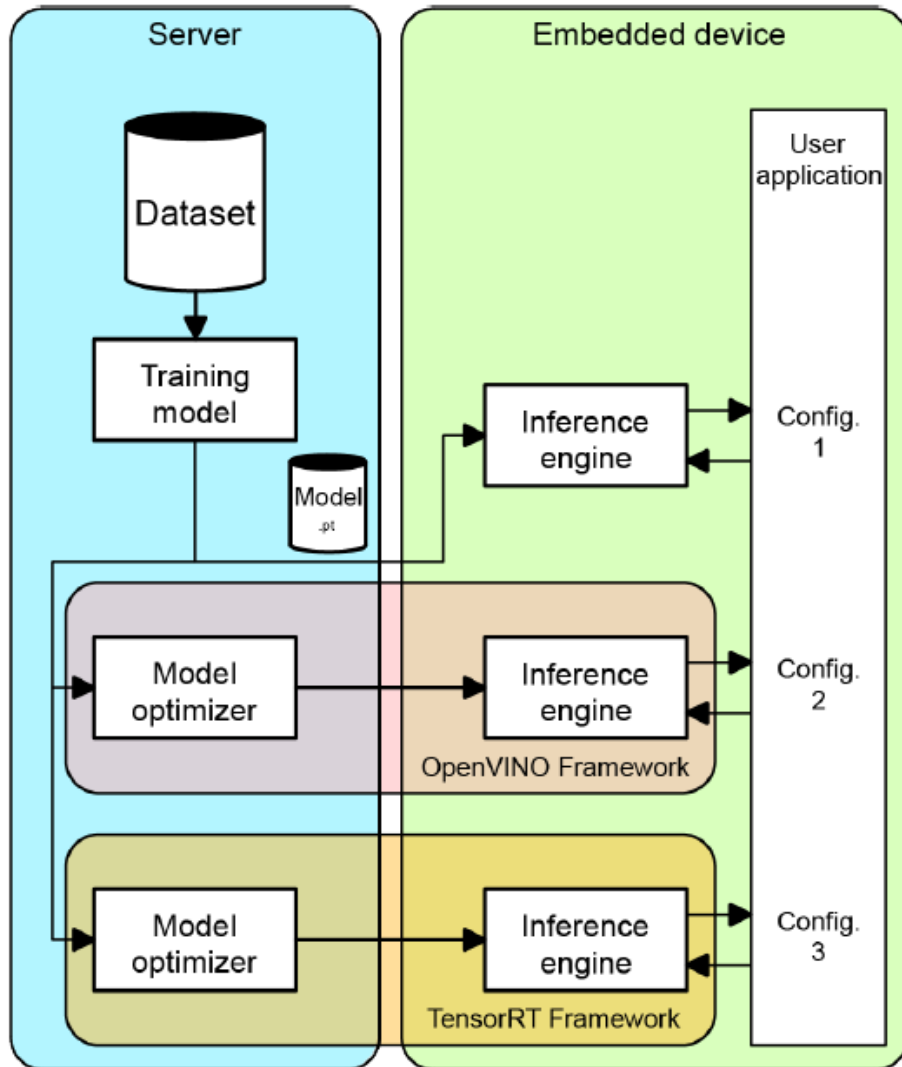
7283 images

# Model Optimization

## 1 of 1 – Workflow of Optimization



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- 1) **Training** of models using PyTorch;
- 2) **Conversion** to an intermediate representation (IR) format via a **Model Optimizer**;
- 3) **Three** separate **deployments** were performed for each model:

- I. PyTorch** → no optimization;
- II. OpenVINO** → optimized for CPU inference;
- III. TensorRT** → optimized for GPU inference.

# Results

## 1 of 3 – Performance comparison: Accuracy

AVERAGE AP@50 VALUES FOR VARIOUS MODELS ON D<sub>3</sub>

| Model   |        | AP@50 (%)   |             |             |             |             |             |
|---------|--------|-------------|-------------|-------------|-------------|-------------|-------------|
|         |        | Head        | Helmet      | Chest       | Vest        | Hand        | Glove       |
| RT-DETR | L      | 99.0        | 98.5        | 97.5        | 99.1        | 92.5        | 88.7        |
|         | X      | 98.3        | 99.0        | 97.4        | <b>99.4</b> | 88.5        | <b>93.7</b> |
| YOLOv11 | Nano   | <b>99.2</b> | <b>99.1</b> | 97.6        | 99.0        | 92.0        | 86.1        |
|         | Small  | 99.1        | 98.6        | <b>97.9</b> | 99.1        | 93.4        | 89.7        |
|         | Medium | 99.1        | 98.9        | 97.6        | 99.2        | 92.8        | 90.7        |
|         | Large  | 99.1        | 98.8        | 97.6        | 99.3        | 92.7        | 90.9        |
|         | XLarge | 99.0        | 99.1        | 97.0        | 99.3        | <b>94.3</b> | 90.5        |
| YOLOv5  | Nano   | 98.0        | 98.7        | 90.1        | 98.7        | 79.9        | 59.3        |
| YOLOv4  | Tiny   | 87.3        | 97.7        | 91.5        | 98.0        | 55.9        | 47.2        |

# Results

## 2 of 3 – Performance comparison: FPS and Disk occupancy

GENERAL OVERVIEW OF THE PERFORMANCE OF THE MODELS, INCLUDING FLOPs, FPS, MAP@50 (CROSS-VALIDATION), MAP@50 (ONLY D<sub>3</sub> AS TEST SET) AND MEMORY OCCUPANCY ON THE DISK.

| Model   |        | MParams | FLOPs<br>(GFlops) | FPS     |          |          | mAP@50           |                   | Weights (MB) |          |          |
|---------|--------|---------|-------------------|---------|----------|----------|------------------|-------------------|--------------|----------|----------|
|         |        |         |                   | Pytorch | OpenVINO | TensorRT | cross-validation | on D <sub>3</sub> | Pytorch      | OpenVINO | TensorRT |
| RT-DETR | L      | 32.8    | 108.0             | 0.8     | 0.2      | - *      | 86.4             | 95.9              | 63.1         | 123.5    | - *      |
|         | X      | 65.4    | 222.5             | 0.5     | 0.1      | - *      | 87.6             | 96.1              | 129.0        | 251.5    | - *      |
| YOLOv11 | Nano   | 2.6     | 6.5               | 6.6     | 2.3      | 10.6     | 86.1             | 95.5              | 5.2          | 10.2     | 12.5     |
|         | Small  | 9.4     | 21.5              | 3.6     | 0.9      | 5.5      | 88.0             | 96.3              | 18.2         | 36.3     | 39.2     |
|         | Medium | 20.1    | 68.0              | 1.7     | 0.3      | 2.3      | 89.2             | 96.4              | 38.6         | 76.8     | 80.7     |
|         | Large  | 25.3    | 86.9              | 1.2     | 0.2      | 1.9      | 89.4             | 96.4              | 63.1         | 123.0    | 101.4    |
|         | XLarge | 56.9    | 194.9             | 0.7     | - **     | - **     | 89.6             | 96.2              | 109.1        | 251.0    | 221.7    |

Note: \* RT-DETR models do not natively support TensorRT optimizations on NVIDIA Jetson Nano. \*\* Insufficient resources for real-time execution.

# Results

## 3 of 3 – Performance comparison: Temperature and Power Consumption

TEMPERATURE AND POWER CONSUMPTION OF DIFFERENT MODELS ON NVIDIA JETSON NANO BOARD

| Model   |        | MParams | Pytorch <sup>+</sup> | Temperature (°C)       |                       | Power Consumption (W) |                        |                        |
|---------|--------|---------|----------------------|------------------------|-----------------------|-----------------------|------------------------|------------------------|
|         |        |         |                      | OpenVINO <sup>++</sup> | TensorRT <sup>+</sup> | Pytorch <sup>+</sup>  | OpenVINO <sup>++</sup> | TensorRT <sup>++</sup> |
| RT-DETR | L      | 32.8    | 72                   | 62                     | -                     | 4.9                   | 3.4                    | -                      |
|         | X      | 65.4    | 72                   | 63                     | -                     | 5.2                   | 3.5                    | -                      |
| YOLOv11 | Nano   | 2.6     | 64                   | 62                     | 60                    | 3.6                   | 2.9                    | 3.0                    |
|         | Small  | 9.4     | 70                   | 64                     | 70                    | 4.3                   | 3.2                    | 4.5                    |
|         | Medium | 20.1    | 72                   | 64                     | 76                    | 4.8                   | 3.4                    | 5.4                    |
|         | Large  | 25.3    | 75                   | 67                     | 80                    | 5.0                   | 3.6                    | 5.6                    |
|         | XLarge | 56.9    | 78                   | -                      | -                     | 5.1                   | -                      | -                      |

**YOLOv11 models**, especially the **Nano and Small variants**, provide a solid balance between **accuracy and real-time performance**, making them suitable for deployment on the **edge device NVIDIA Jetson nano**.

| Framework | Pros                                        | Cons                                          |
|-----------|---------------------------------------------|-----------------------------------------------|
| PyTorch   | Easy to use and flexible for training       | Not optimized for edge devices                |
| OpenVINO  | CPU efficient, low consumption              | Lower speed, not suitable for rapid inference |
| TensorRT  | Very high speed on GPU, ideal for real-time | Higher energy consumption with large models   |

 Apply quantization and pruning to reduce model size and improve efficiency

 Explore multi-camera and multi-angle setups for wider scene coverage

 Extend detection capabilities to unseen or custom PPE types

 Improve domain adaptation for deployment in diverse industrial environments

 Investigate on-device learning or model updating for continuous adaptation



# Any questions?

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